REIMAGINE CANDY BOWL
“MINI CANDY CLAW MACHINE”

Team name: THREE QUARTERS

DATE: 12/02/2025

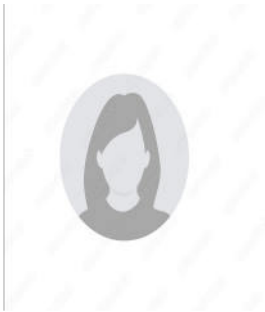
Team members:

Long

Kit

Sofi

Client:



UW Graduate Academic Counselor

Project Overview & Client Need

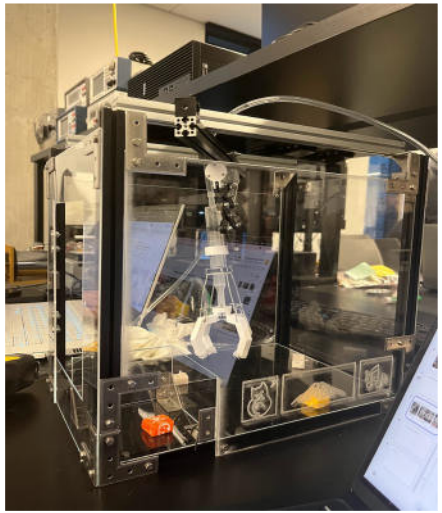
Project Overview:

Claw machine candy bowl that features 2-axis sliding system for positioning and hydraulic-powered claw for gripping candy.



Client Need:

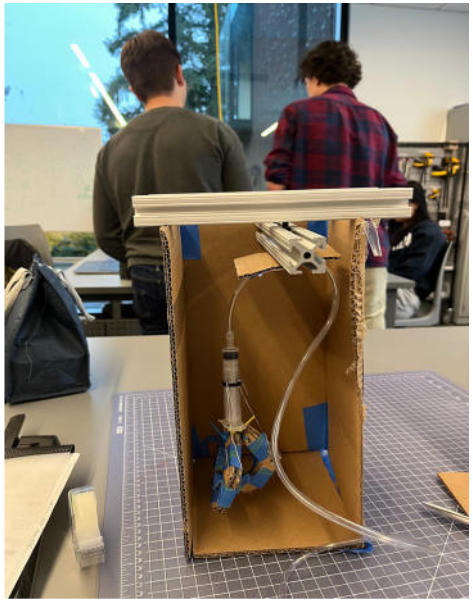
Jane wants more students to visit her space and interact. The candy bowl encourages students to engage and spark conversation.

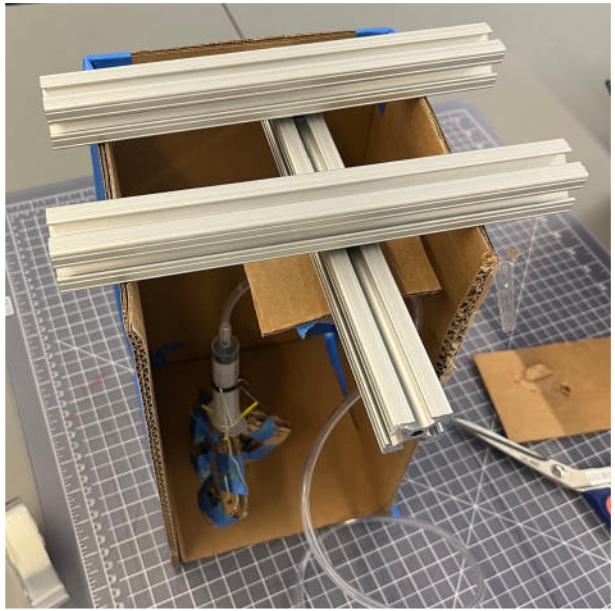


User Requirements & Engineering Specifications

User Requirements	Engineering Specification	Actual Design:
Compact and Portable	About 12x12x12 inches Weight < 5lbs	12x12.5x12.8 inches Around 3lbs
Cats, nature, music vibe Visual appealing	50% students comment on the design	Engraved images of cats, music, nature Tree decoration by wood
Stable	Stationary while being used	Stabilized by makerbeams structure
Durable	No break/deform after being transported/used	Stabilized by makerbeams structure
Cost-efficient	Cost < $150	$126.70

Design Process & Concept Exploration





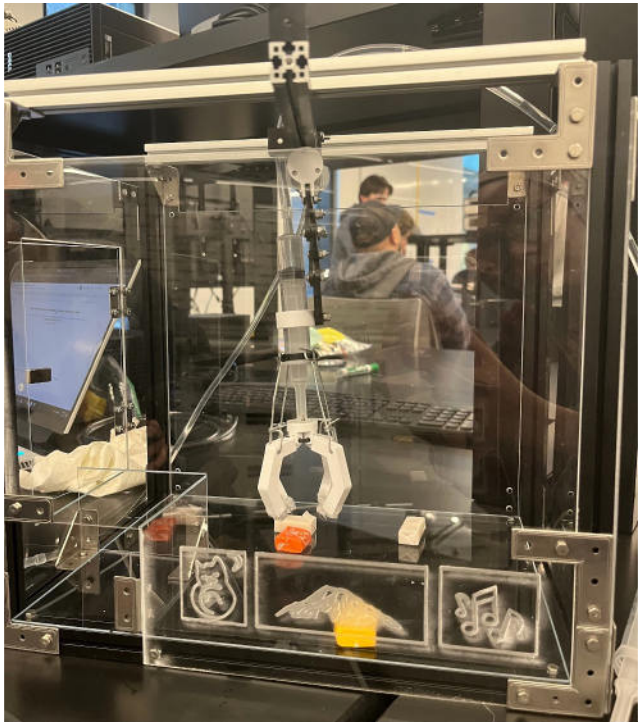


Week 4 - 5

Week 7

Most up to date design (CAD models, diagrams, visuals)

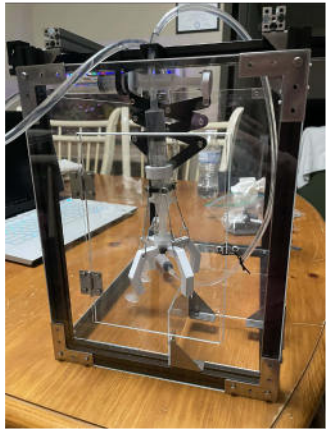
- Hydraulic powered claw
- Transparent acrylic enclosure for full visibility
- Side refill door for easy loading
- Small opening at the corner for user to drop the candy and grab the candy retrieved

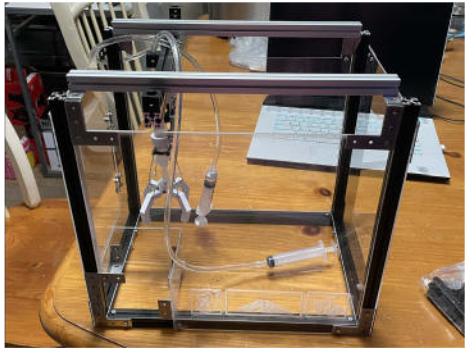


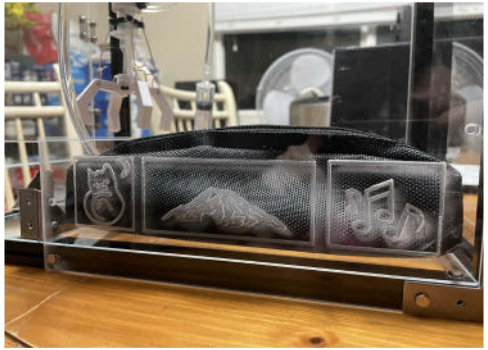
Prototype Build & Testing











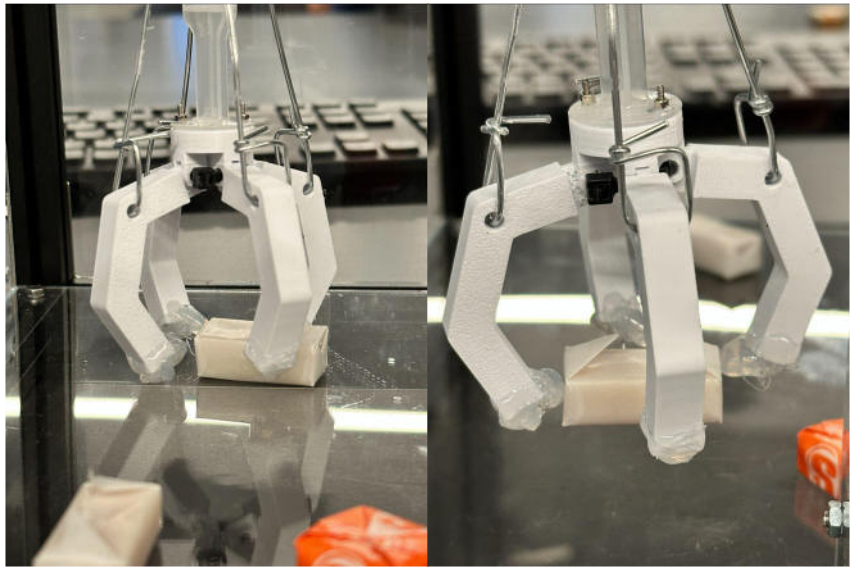
Testing Video:



Limitations & Future Improvements



Screws Block X-direction movement



Precise positioning needed to pick up candy



Claw gets stuck on drop

Final Recommendations

- Larger Box to make Z-direction movement easier
- Redesign Wire for claw (Identical length)
- Mount for syringes

Value Delivered

- Good Ice Breaker
- Fun and Interactive
- Singularize Candies



Acknowledgments

- Client: Jane
- Professor: Eli and Sirine
- TA: Masha
- Laser cut assistant: Hailey, Brenda
- Friends

APPENDIX A: Cost

Material + Cost	Price (in dollar): P	Quantity: Q	Cost: C = P*Q	
300mm makerbeam	4	9	$36.00	
150mm makerbeam	2.5	2	$5.00	
40mm makerbeam	1	2	$2.00	
Acrylic sheet	5	6	$30.00	
3D printing	5	1.5	$7.50	cost/hr
Syringe	0.75	4	$3.00	
Air tube	0.25	6	$1.50	cost/ft
Magnet	X	2	$0.00	
Paper clip	X	4	$0.00	
Right Angle Bracket (L)	1	16	$16.00	
Triangular Inner Bracket	1.5	7	$10.50	
Straight Bracket	1	2	$2.00	
M3 Hexagon Bolt	0.15	54	$8.10	
Button Head Hex Bolt	0.1	10	$1.00	
M3 Nut	0.05	70	$3.50	
Total			$126.10	